



COMSATS University, Islamabad.
Department of Computer Science

Class: BCT VII

Subject: CSC418 - DevOps and Cloud Computing

Instructor: Dr. Muhammad Imran

[CLO4]: Apply DevOps pipeline automation techniques for code deployment.

Submission Date: 16-12-25

Marks: 10

Lab Assignment 4

The CYBER-DEF25 Challenge requires participants to submit their trained AI-based Malware Detection model as an inference container. The container is Docker-based and must encapsulate the following:

- The trained cybersecurity model (model.pkl).
 - The inference script that loads the model, analyzes incoming network logs, detects potential threats, and outputs results (inference.py).
 - All dependencies and environment setup needed for reproducibility (requirements.txt)
 - inference.py expects log files to be placed in /input/logs inside the container and saves detection results (alerts.csv) in the /output folder.
- a) Create a Docker image using Dockerfile and push that image to the DockerHub.
 - b) Write YAML manifest for creating Deployment to run the malware detection container over the Kubernetes cluster. **[3]**
 - c) Write YAML manifest for creating Service to handle the incoming traffic automatically.. **[3]**
 - d) Write YAML manifest for creating Persistent Volume Claim (PVC) to persist input logs and generated alerts. **[4]**

What to submit

It's an individual assignment and you need to submit a well formatted report describing your application, documenting all the micro steps, along with the screenshots you follow for completing the assignment. Furthermore, you need to provide in the report your deployment, service and PVC YAML files.